Asterline Assistant ? Take-home submission

Repository: https://github.com/satvik310706/asterline-assistant
LangSmith traces: [PASTESHAREDTRACELINKSHERE]

Executive summary
A research assistant that answers only from an immutable corpus, cites evidence,
handles supersession and abstention, and remembers safe user context across restarts.
The baseline is frozen; the improved configuration adds seven typed agent stages.

Reproduction
uv sync
uv run pytest -q
uv run asterline ask "What is the current Launch ingestion rate?" --provider fake
uv run asterline run

Generated evaluation artifacts
baselinefake.json: provider=fake config=baseline cases=80
  model: deterministic
  fallback disabled: True
  retrievalrecallat5: 0.9743589743589743 (78/80 applicable)
  mrr: 0.7653846153846156 (78/80 applicable)
  factaccuracy: 0.725 (80/80 applicable)
  citationvalidity: 1.0 (80/80 applicable)
  citationprecision: 0.5961538461538461 (78/80 applicable)
  citationrecall: 0.8076923076923077 (78/80 applicable)
  conflicthandling: 0.42857142857142855 (7/80 applicable)
  abstentionaccuracy: 0.9125 (80/80 applicable)
  providererrorcount: 0
  evalerrorcount: 0
  llmcallsperquestion: 1.0
  fallbackfraction: 0.0 (80/80 applicable)
baselinegemini.json: provider=gemini config=baseline cases=80
  model: gemini-3.5-flash
  fallback disabled: True
  retrievalrecallat5: 0.95 (80/80 applicable)
  mrr: 0.8425925925925927 (18/80 applicable)
  factaccuracy: 1.0 (18/80 applicable)
  citationvalidity: 1.0 (18/80 applicable)
  citationprecision: 0.9074074074074073 (18/80 applicable)
  citationrecall: 1.0 (18/80 applicable)
  conflicthandling: 0.0 (0/80 applicable)
  abstentionaccuracy: 1.0 (18/80 applicable)
  providererrorcount: 62
  evalerrorcount: 0
  llmcallsperquestion: 1.5625

fallbackfraction: 0.0 (18/80 applicable)

baselineollama.json: provider=ollama config=baseline cases=80

model: llama3.2:3b

fallback disabled: True

retrievalrecallat5: 0.9743589743589743 (78/80 applicable)

mrr: 0.7653846153846156 (78/80 applicable)

factaccuracy: 0.4625 (80/80 applicable)

citationvalidity: 1.0 (80/80 applicable)

citationprecision: 0.6132478632478632 (78/80 applicable)

citationrecall: 0.46794871794871806 (78/80 applicable)

conflicthandling: 0.5714285714285714 (7/80 applicable)

abstentionaccuracy: 0.7875 (80/80 applicable)

providererrorcount: 0

evalerrorcount: 0

llmcallsperquestion: 1.3875

fallbackfraction: 0.0 (74/80 applicable)

improvedfake.json: provider=fake config=improved cases=80

model: deterministic

fallback disabled: True

retrievalrecallat5: 0.9358974358974359 (78/80 applicable)

mrr: 0.7049145299145301 (78/80 applicable)

factaccuracy: 0.775 (80/80 applicable)

citationvalidity: 0.975 (80/80 applicable)

citationprecision: 0.6217948717948718 (78/80 applicable)

citationrecall: 0.7713675213675213 (78/80 applicable)

conflicthandling: 1.0 (7/80 applicable)

abstentionaccuracy: 0.9125 (80/80 applicable)

providererrorcount: 0

evalerrorcount: 0

llmcallsperquestion: 0.6875

fallbackfraction: 0.0 (80/80 applicable)

improvedollama.json: provider=ollama config=improved cases=80

model: llama3.2:3b

fallback disabled: True

retrievalrecallat5: 0.9487179487179487 (78/80 applicable)

mrr: 0.7017094017094019 (78/80 applicable)

factaccuracy: 0.55 (80/80 applicable)

citationvalidity: 0.975 (80/80 applicable)

citationprecision: 0.6089743589743589 (78/80 applicable)

citationrecall: 0.5299145299145299 (78/80 applicable)

conflicthandling: 0.8571428571428571 (7/80 applicable)

abstentionaccuracy: 0.8125 (80/80 applicable)

providererrorcount: 0

evalerrorcount: 0

llmcallsperquestion: 1.2875

fallbackfraction: 0.0 (71/80 applicable)

memoryfake.json: provider=fake scenarios=34 passed=34/34
  memorywriteprecision: 1.0 (aggregate over 34 scenarios)
  memorywriterecall: 1.0 (aggregate over 34 scenarios)
  forgetsuccess: 1.0 (aggregate over 34 scenarios)
  isolationleakage: 0 (aggregate over 34 scenarios)
  secretleakcount: 0 (aggregate over 34 scenarios)
  sensitivestoredcount: 0 (aggregate over 34 scenarios)
  correction-company pass rate: 1.0
  correction-self pass rate: 1.0
  forget pass rate: 1.0
  injection pass rate: 1.0
  isolation pass rate: 1.0
  list-memory pass rate: 1.0
  memory-not-evidence pass rate: 1.0
  persistence pass rate: 1.0
  preference pass rate: 1.0
  secret pass rate: 1.0
  sensitive pass rate: 1.0
  stale-confirmation pass rate: 1.0

Evidence and memory guarantees
Citations are parsed and validated against retrieved chunks. Memory is labelled unverified context and never counts as company evidence. Secrets and sensitive personal data are redacted or rejected before storage and tracing. SQLite is the source of truth; hard delete removes memory rows, vectors, sessions, and messages.

Provider resilience smoke
A real Gemini request returned HTTP 429 (quota exhausted); the app recorded the sanitized primary error and successfully served the answer through the locally installed Ollama llama3.2:3b model with a citation validated against retrieved evidence. This is a resilience smoke test, not a Gemini answer-quality evaluation.

Limitations
FakeLLM reports verify plumbing, not provider quality. Gemini quality runs and remote LangSmith links require the author’s credentials and quota. Deleted data may remain in already-shipped remote traces. Regex matching is a rough metric.

AI use disclosure
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- Codex (GPT) wrote most code under my direction.
- Claude helped with planning, datasets, and prompts.
- Tests and evaluation runners were used to verify behavior and expose failures.

Author edits

Add any additional tools, prompts, or review details here before submission.

Design excerpt
Design

mermaid
flowchart LR
  Q[User question] --> R[router]
  R --> M[memoryreader\nUNVERIFIED USER CONTEXT]
  R --> S[retriever\nhybrid dense + BM25]
  S --> A[evidenceanalyst\ncurrent/superseded]
  M --> W[answerwriter]
  A --> W
  W --> V[verifier]
  V --> MW[memorywriter\npolicy gate]
  MW --> DB[(SQLite user memory)]
  A --> C[(Chroma corpus index)]

Baseline keeps the original route ? retrieve ? answer path frozen. Improved uses seven named graph nodes. Corp

Trade-offs: deterministic policy and validation are easier to explain than a fully autonomous agent, but FakeL